VECTOR



SOME/IP with CANoe Communication Setup

Part 1. Introduction

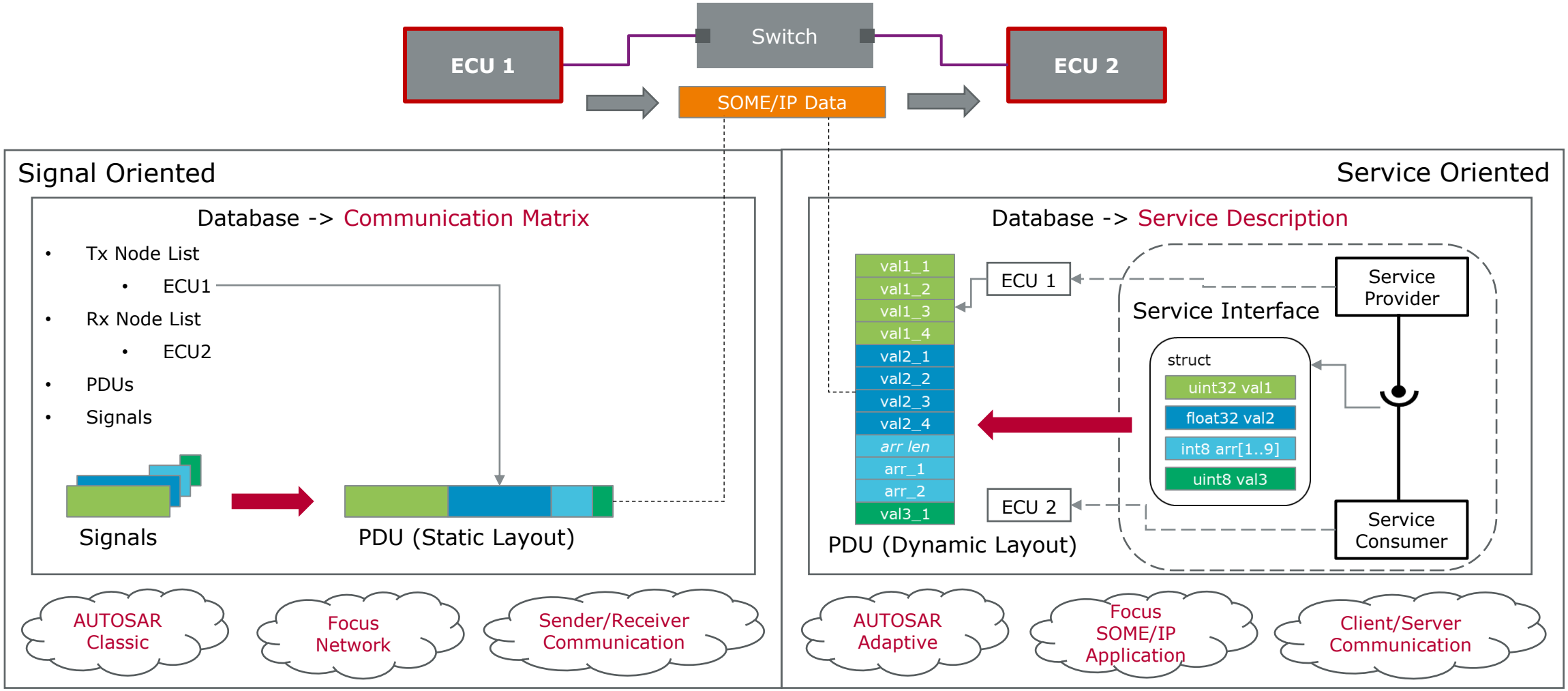
V1.0 | 2023-07-18

Agenda

- **Overview**
- Communication Setup
- Technical References
- Summary

2 perspectives on SOME/IP communication

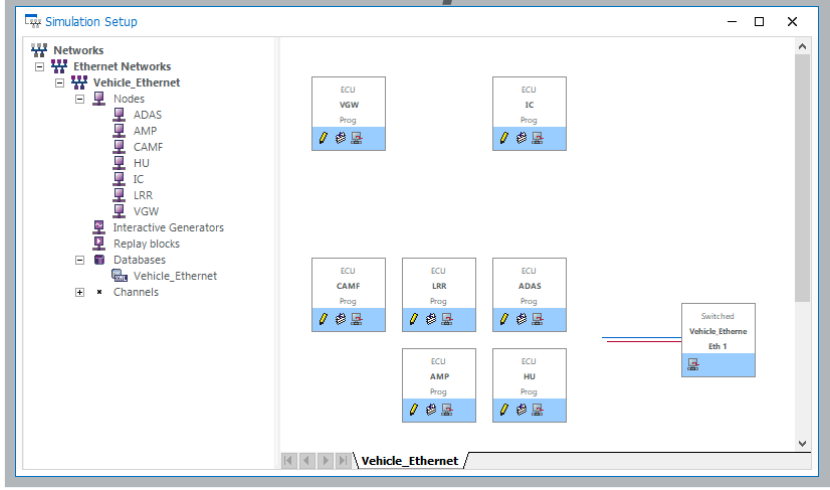
- Let's assume that 2 ECUs communicate via SOME/IP.



SOME/IP Support in CANoe

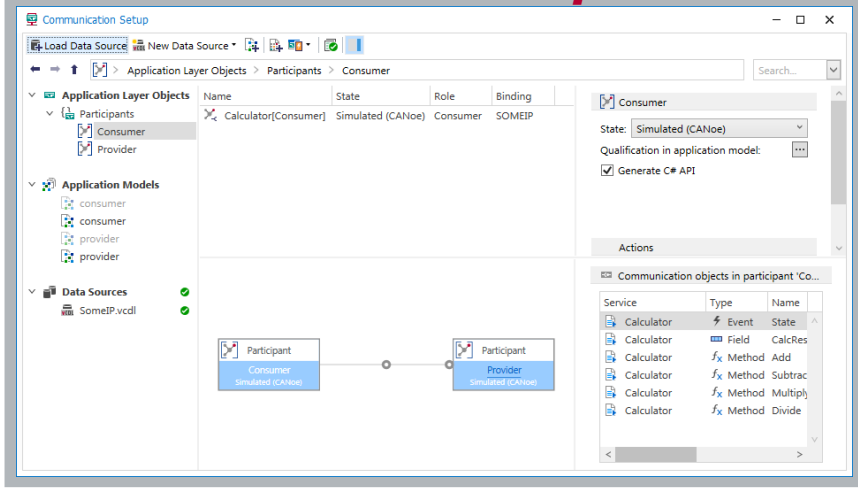
CANoe

Simulation Setup



Communication Setup

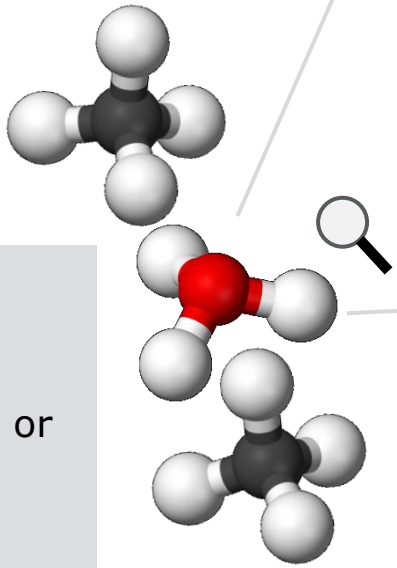
Focus !!



Communication	Signal Oriented Communication	Service Oriented Communication
Database	ARXML(AUTOSAR Classic)	ARXML(AUTOSAR Adaptive), vCDL
Configuration	AUTOSAR PDU, Node, Network	Service, Participant, Binding
Feature Focus	Protocol	Application
Data types	Simple (layout predefined)	Complex (runtime serialization)
Abstraction	Low (signal / PDU)	Higher (service interface)
Use Case	Rest Bus Simulation, Low-level Fault Test	Rest Bus Simulation, Application Test

SOME/IP in Communication Setup

- Service Communication Object is defined by
  - Service interface type
    - Method
    - Event
    - Field
  - End points
    - Provider / Consumer
    - Switched to simulated or real



```
interface Calculator {

  method float Add(int, int);

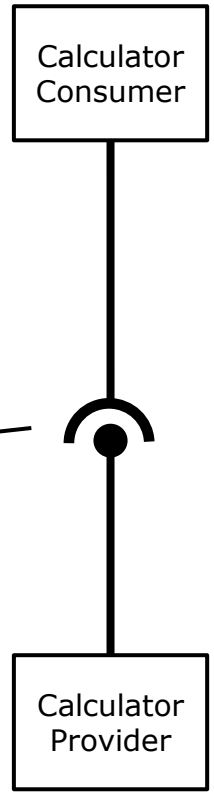
  method float Substract(int, int);

  method float Multiply(int, int);

  method float Divide(int, int);

  field  int32 CalcResult {get;set;notify;}

   event struct State {
     int AddCount;
     int SubstractCount;
     int MultiplyCount;
     int DivideCount;
  }
 }
```



Any pair of endpoints may:

… connect distinct physical devices, or

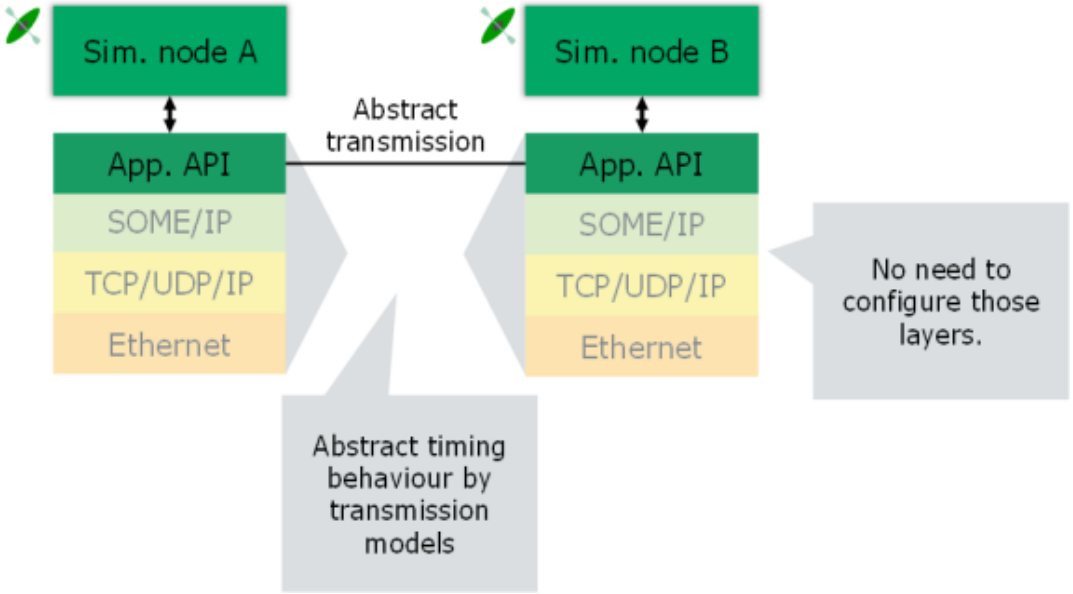
… share a MAC address on the same PHY, or

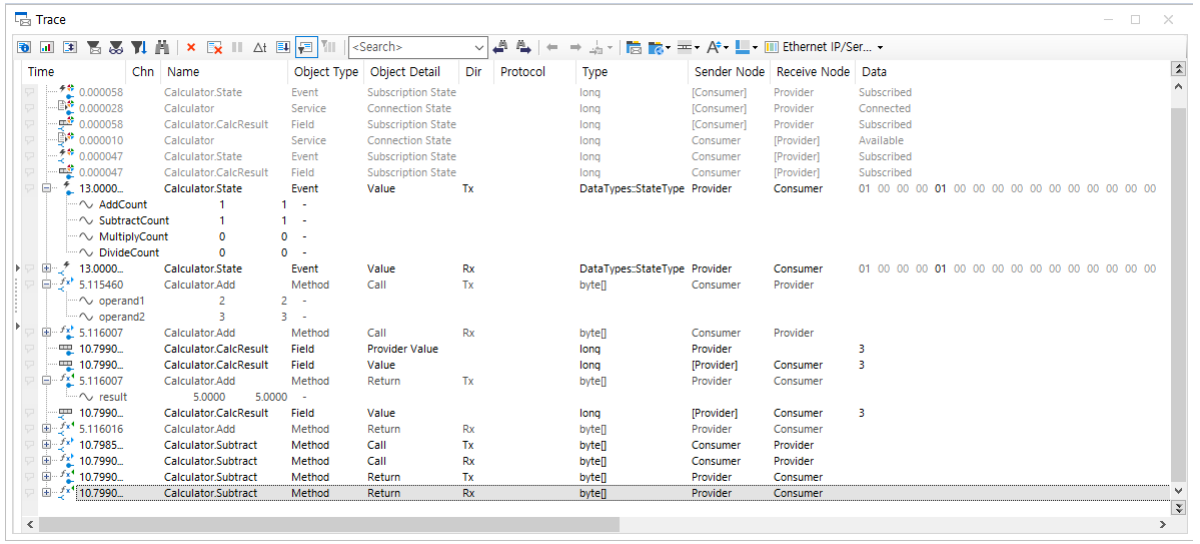
… share an IP address, or

… connect abstract simulation code.

SOME/IP in Communication Setup

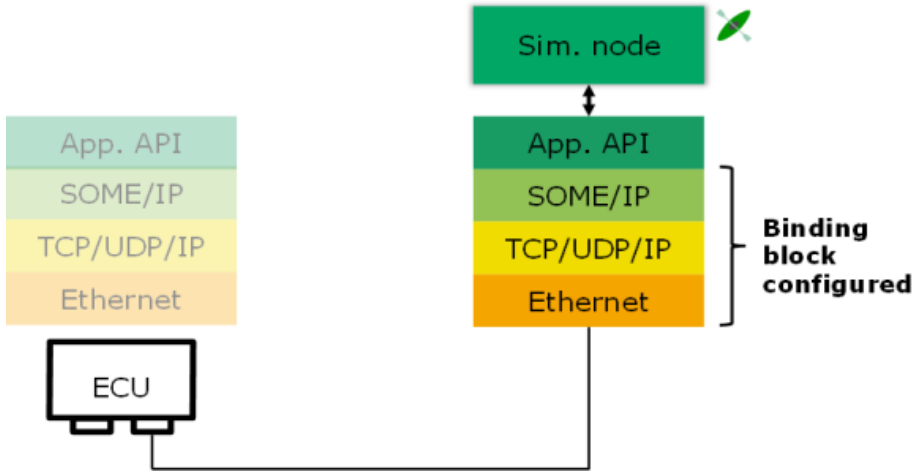
- Used to configure transmission between the endpoints of a Communication Object
- Determine which protocol and network must be used for transmission
- Abstract Binding
  - Enables transmission between simulated endpoints within CANoe
  - Well suited for Virtual prototyping
  - Communication Object is directly linked with Abstract Binding

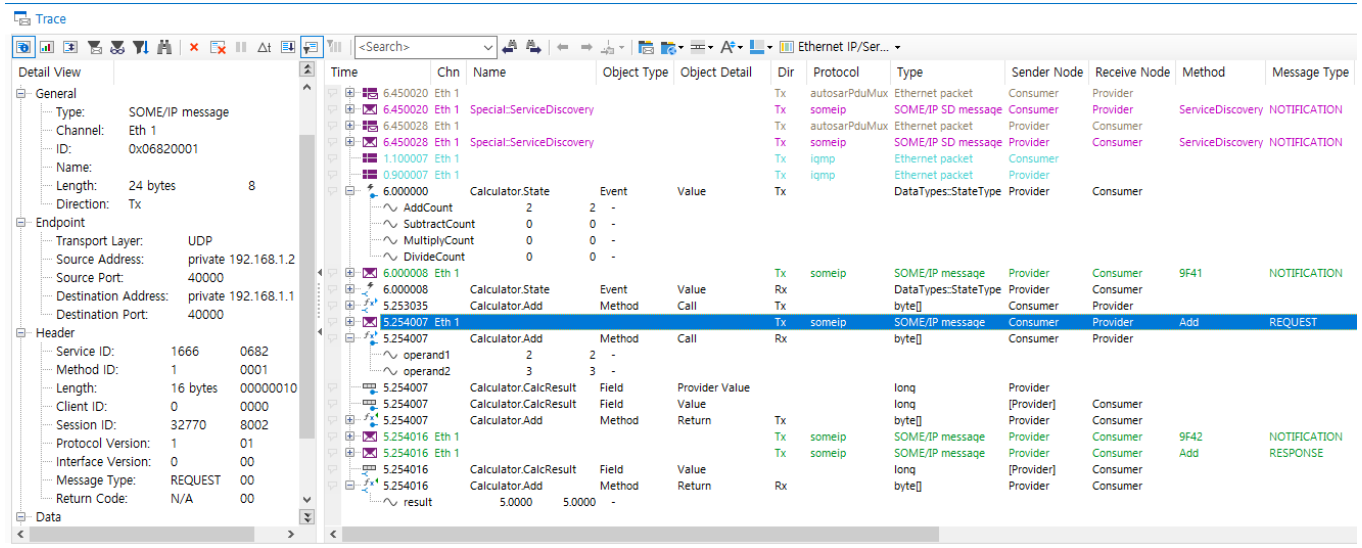




SOME/IP in Communication Setup

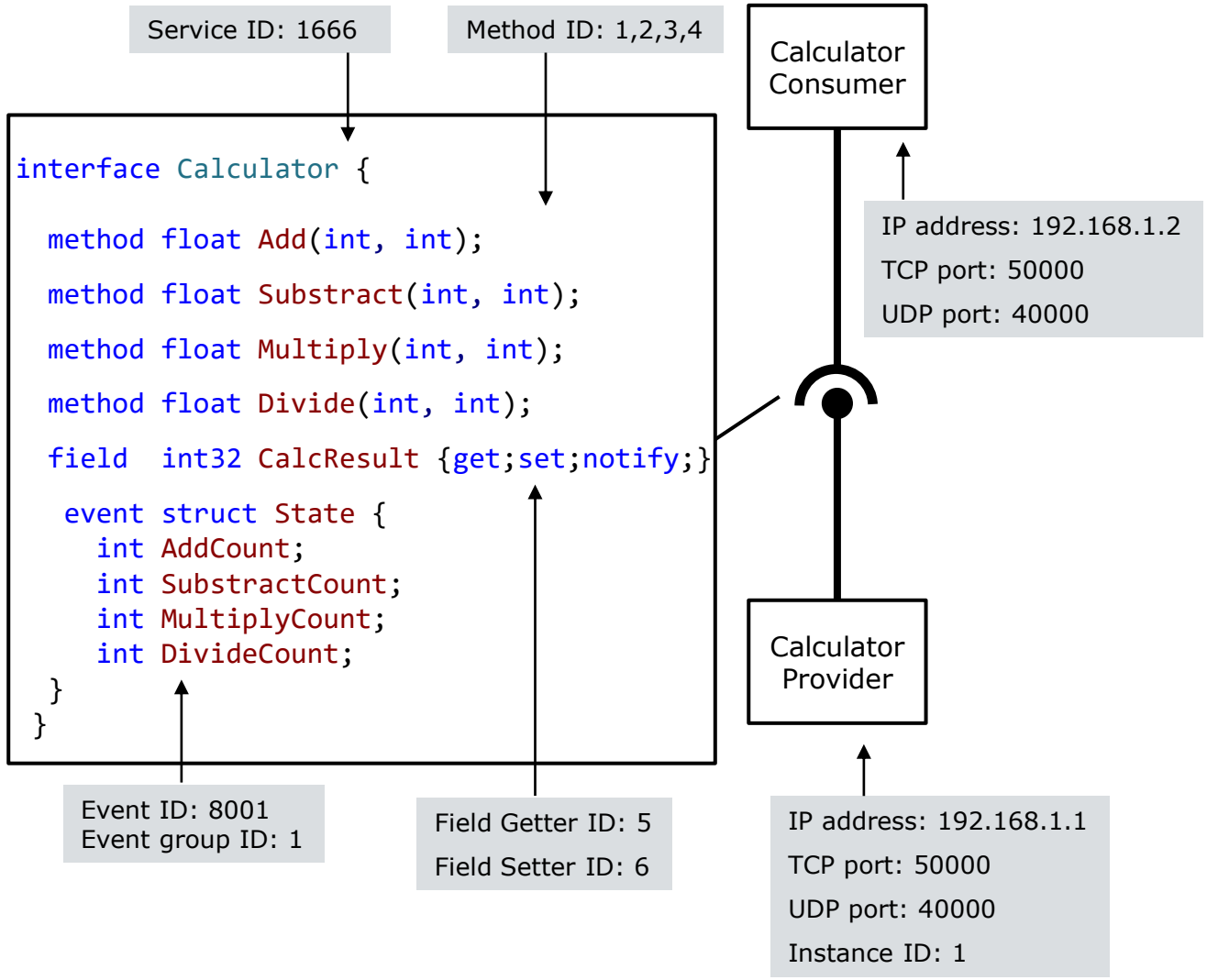
- SOME/IP Binding
  - Enables transmission between endpoints base on SOME/IP via TCP/UDP/IP and Ethernet
  - Supports SOME/IP, SOME/IP Service Discovery protocol
  - Communication Object can be linked with SOME/IP Binding through Abstract Binding





SOME/IP in Communication Setup

- SOME/IP Binding
  - Service Interface Mapping required
    - Service ID
    - Method ID
    - Event ID
    - Event Group
  - SOME/IP Endpoint Mapping required
    - IP Address
    - UDP/TCP Port Number
    - Provider Instance ID



VECTOR

Terms in CANoe

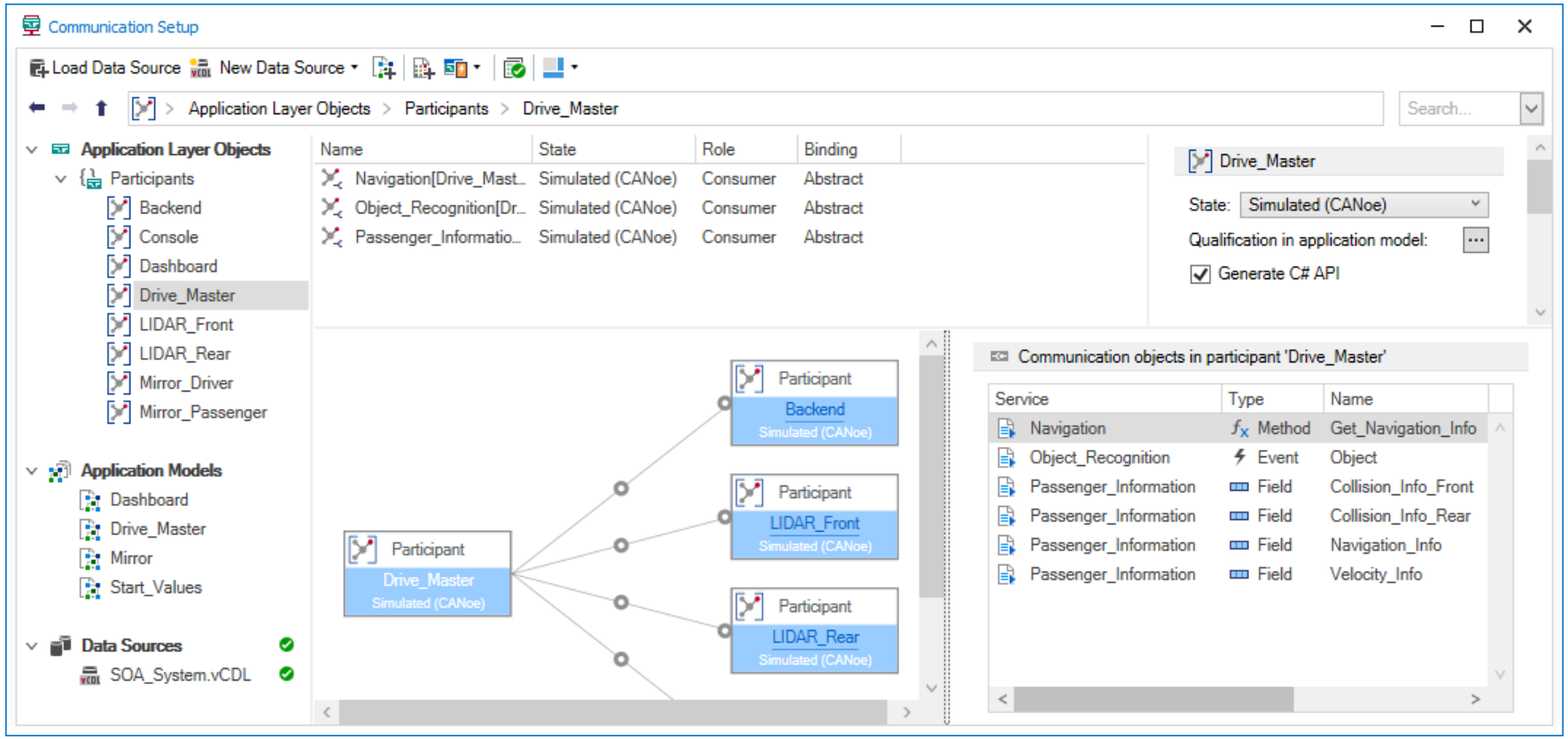
Simulation Setup	Communication Setup	Description
Database	Data Source	*The data in the data source is not automatically used by CANoe. To enable the use of participants/endpoints in CANoe (e.g. for simulation, analysis, display in the Symbol Explorer), you must first configure them in the Communication Setup (integrate into CANoe).*
-	Data Source - vCDL	*The acronym vCDL stands for Vector Communication Description Language and is a domain-specific language (DSLClosed) for the description of communication objects in CANoe.*
ECU/Node	Participants	*Participants/distributed objects can be used to group communication endpoints that are logically associated; e.g. because they belong to the same ECU, or because they are implemented by the same software component. Thus the participants concept allows a more flexible modeling.*
Frame/PDU/Signal	Communication Object – Service	*Communication objects abstract communication of different levels and are not bus-specific. However, frames on the respective bus system can still be analyzed.*
Interaction layer	Binding	*The binding replaces the interaction layer concept. It links abstract communication objects and distributed objects to concrete bus systems and protocols.*
Simulation Node	Application Model	*Application Logic is implemented as a Application Model via CAPL programming*

Agenda

- Overview
- **Communication Setup**
- Technical References
- Summary

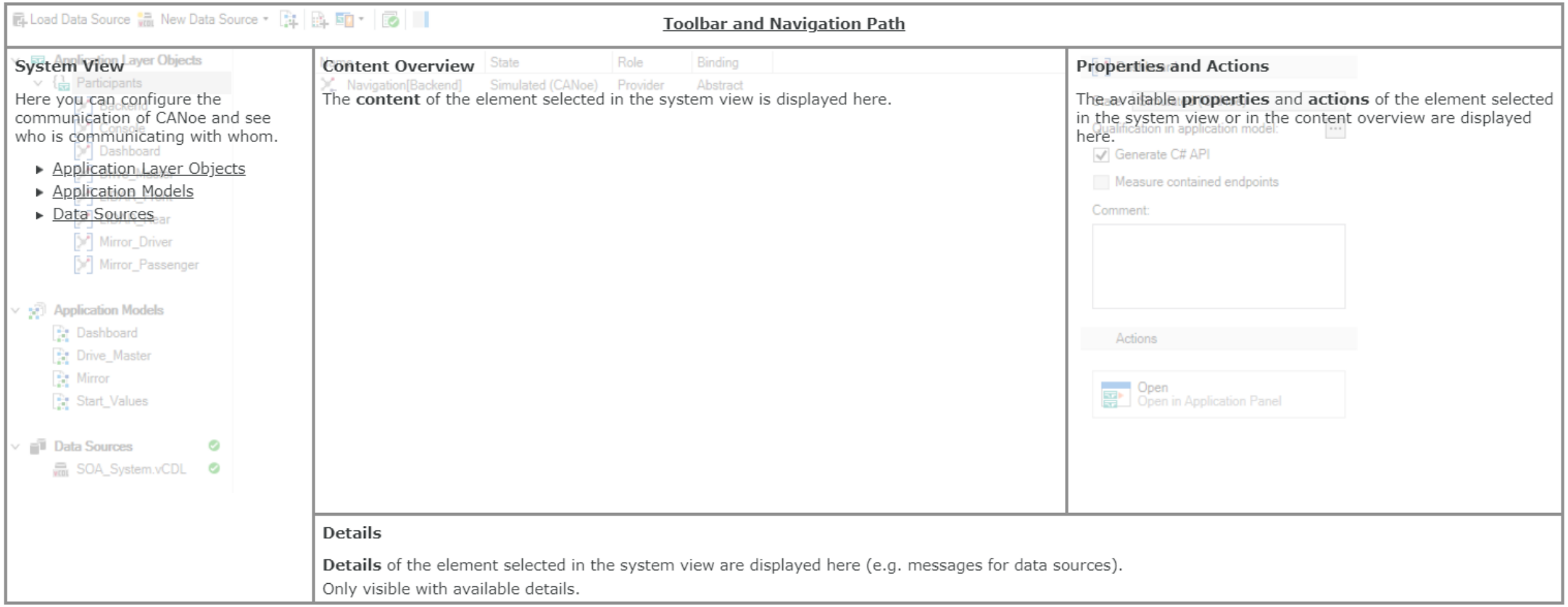
Overview

- The Communication Setup is the central starting point for communication configuration.
  - the application layer (application models)
  - the communication layer (communication, bindings)
  - the transmission medium (hardware, bindings)



Window Layout

- System View
  - Application Layer Objects : Display ‘Namespaces – Participants – Services’
  - Application Models : Programming ‘CAPL’ or ‘C#’
  - Data Sources : ‘vCDL’ which includes SOME/IP configuration



Workflow Example

- Considered application scenario
  - Service interface: *Calculator*
  - Service Provider: *Provider*
  - Service Consumer: *Consumer*

- Input File
  - Data Source: *SomeIP.vcdl*
    - Definition of interface: *Calculator*
    - SOME/IP communication parameters
    - Bindings
  - Application Model: *provider.can*, *consumer.can*
    - Service logics : Four arithmetic
    - Notify Event(State) and Field(Result Value)

- Following workflows are shown next
  - Workflow 1 – Pure simulation
    - **Both** *Provider* **and** *Consumer* are simulated in CANoe
  - Workflow 2 – Partial Simulation with a real target
    - **Either** *Provider* **or** *Consumer* is simulated in CANoe

```
interface Calculator {

  method float Add(int, int);

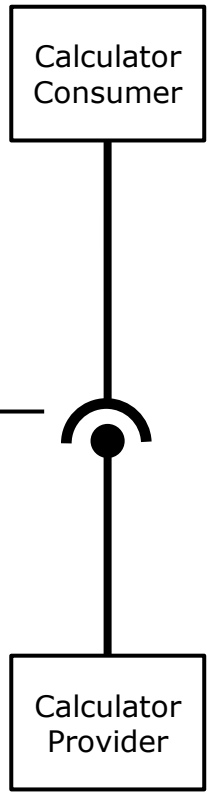
  method float Substract(int, int);

  method float Multiply(int, int);

  method float Divide(int, int);

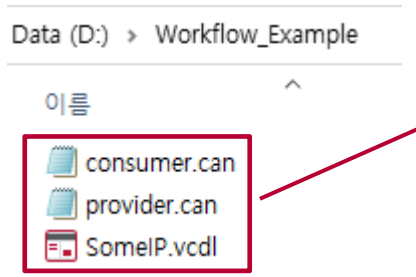
  field  int32 CalcResult {get;set;notify;}

  event struct State {
     int AddCount;
     int SubstractCount;
     int MultiplyCount;
     int DivideCount;
  }
}
```



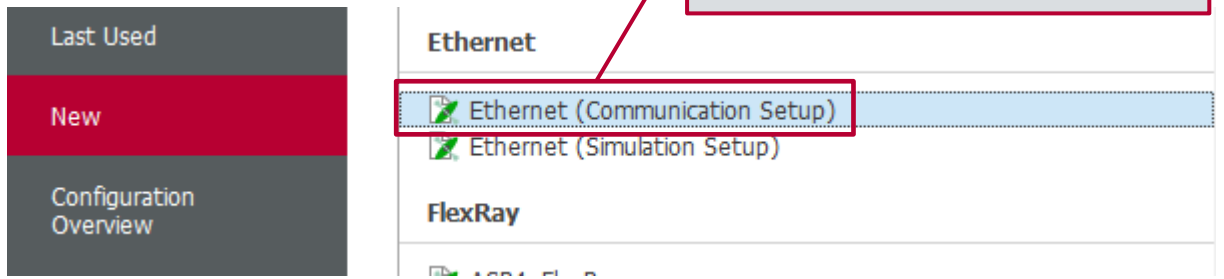
Download Input Files Download Solution

Workflow 1 - Preparation



Step 0

Prepare Sample vCDL, CAPL files -> Download

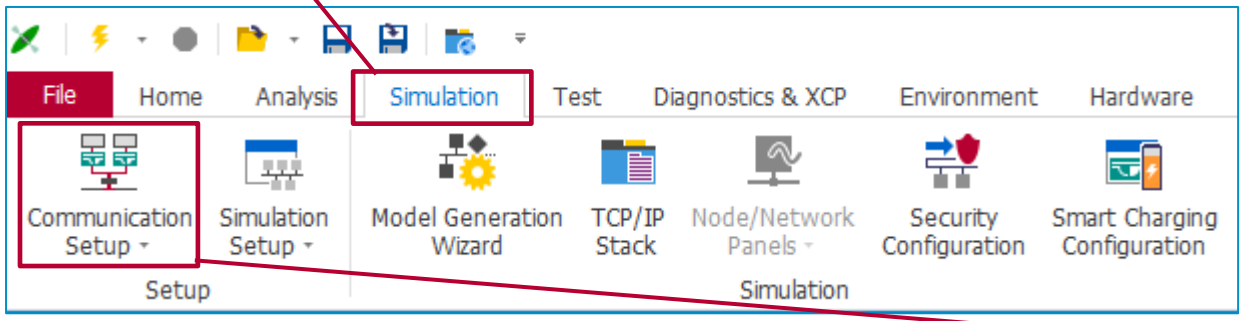


Step 1

Start by creating a new Ethernet configuration

Step 2

Switch to tab *Simulation*



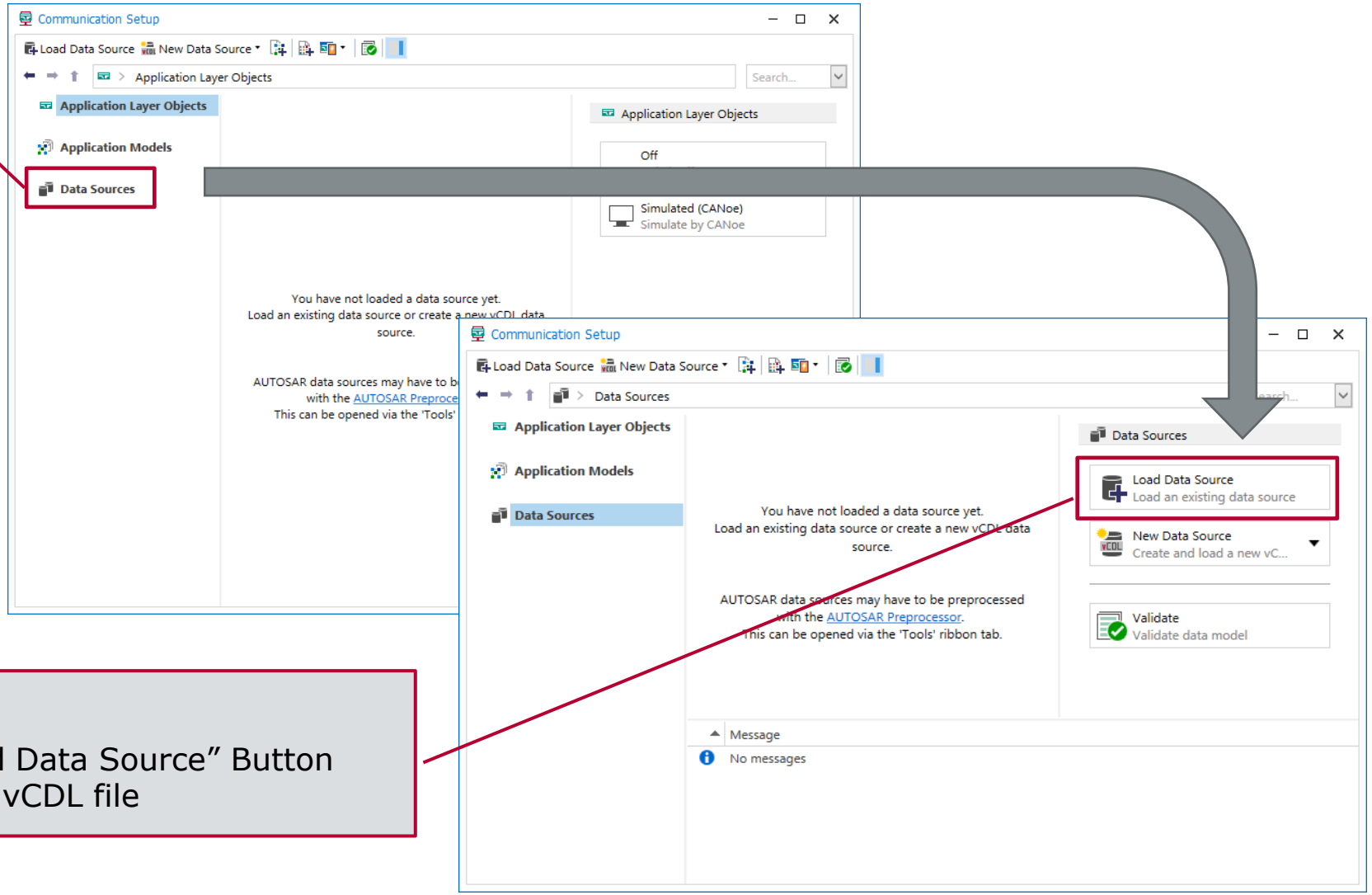
Step 3

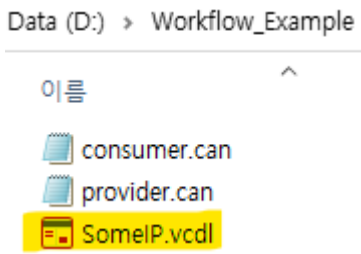
Open dialog by clicking *Communication Setup*

Workflow 1 – Import Data Source

Step 4

Click *Data Sources* to open the import view





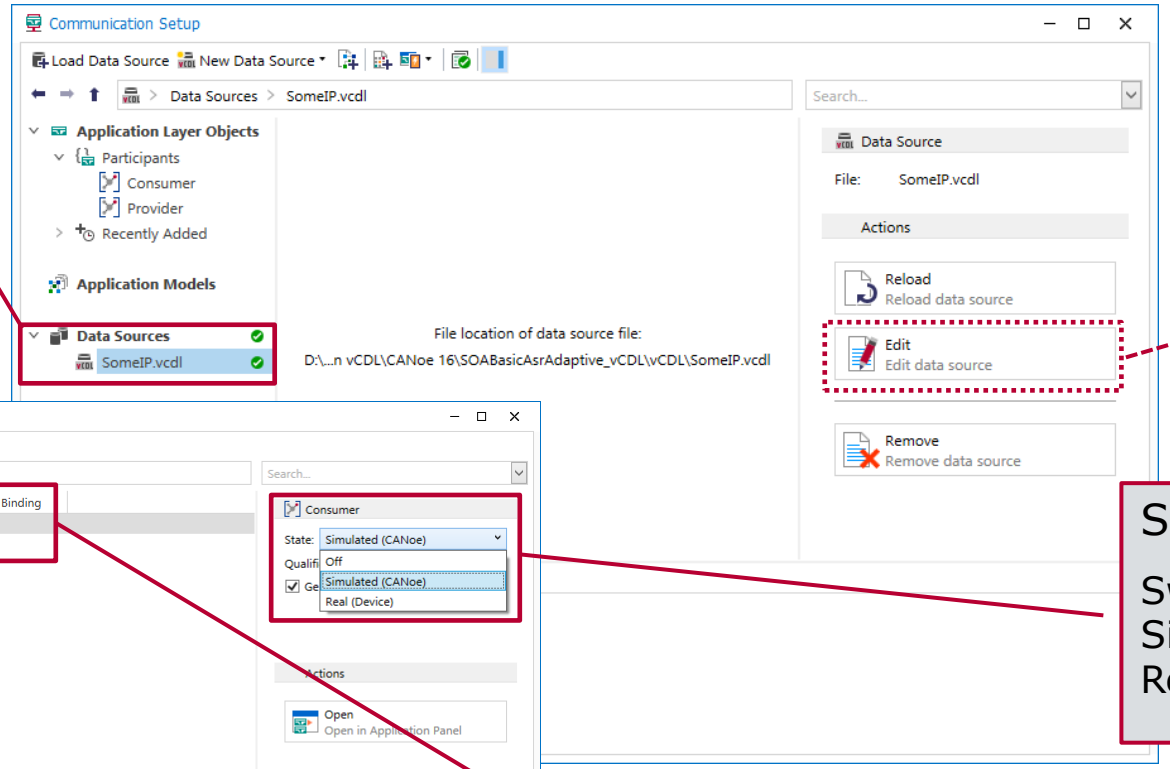
Step 5

Click „Load Data Source” Button and select vCDL file

Workflow 1 – Check Import Result

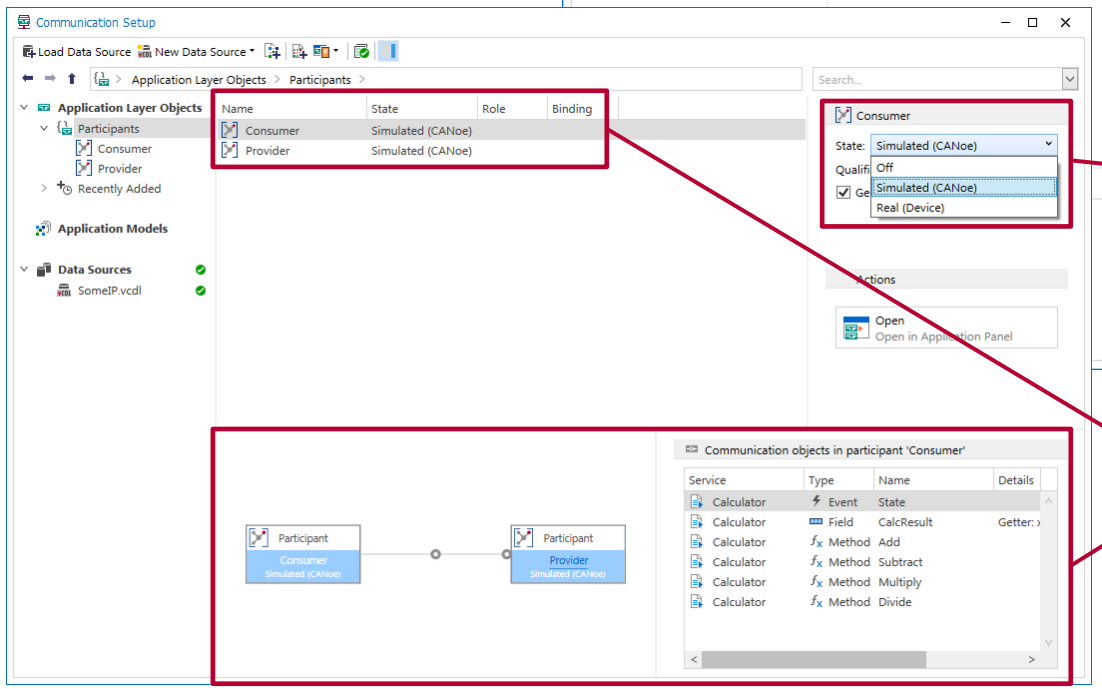
Step 6

A successful import is indicated by the green button



Step 7 (optional)

Imported services and data types can be modified via this Button



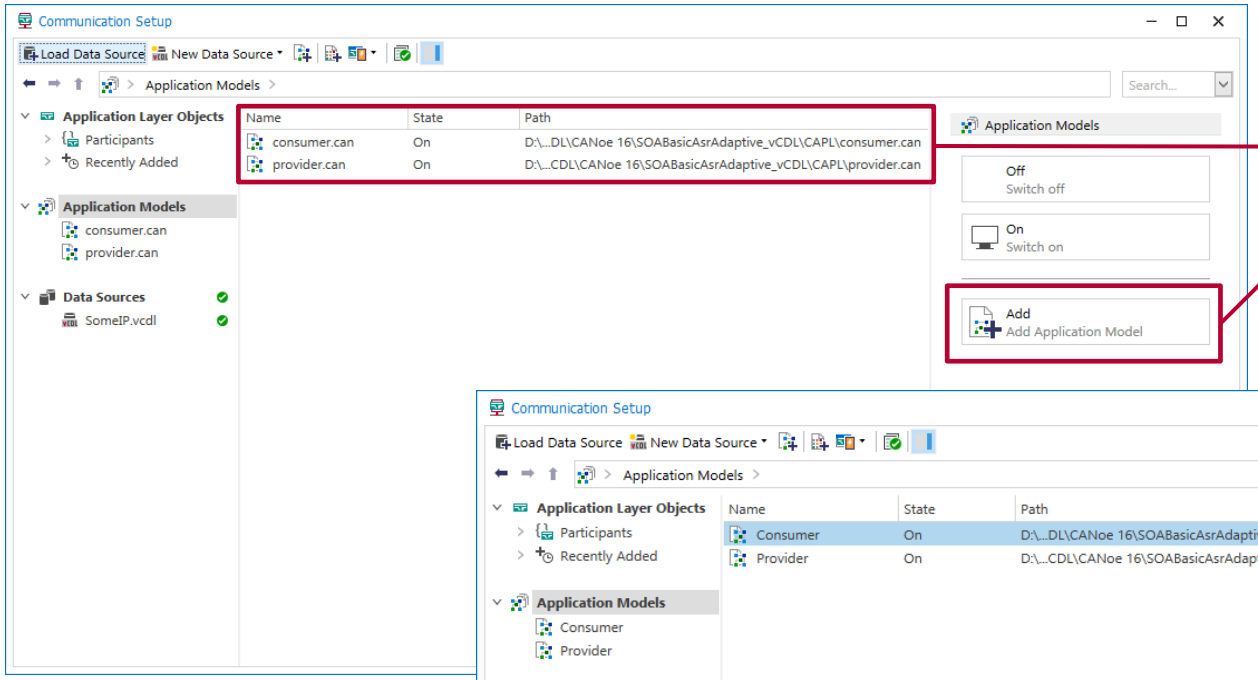
Step 9

Switch Participant to Simulated(CANoe) or Real(Device) Mode

Step 8

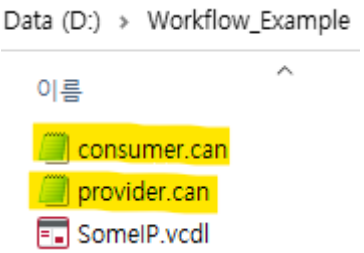
The imported elements are shown in a Details

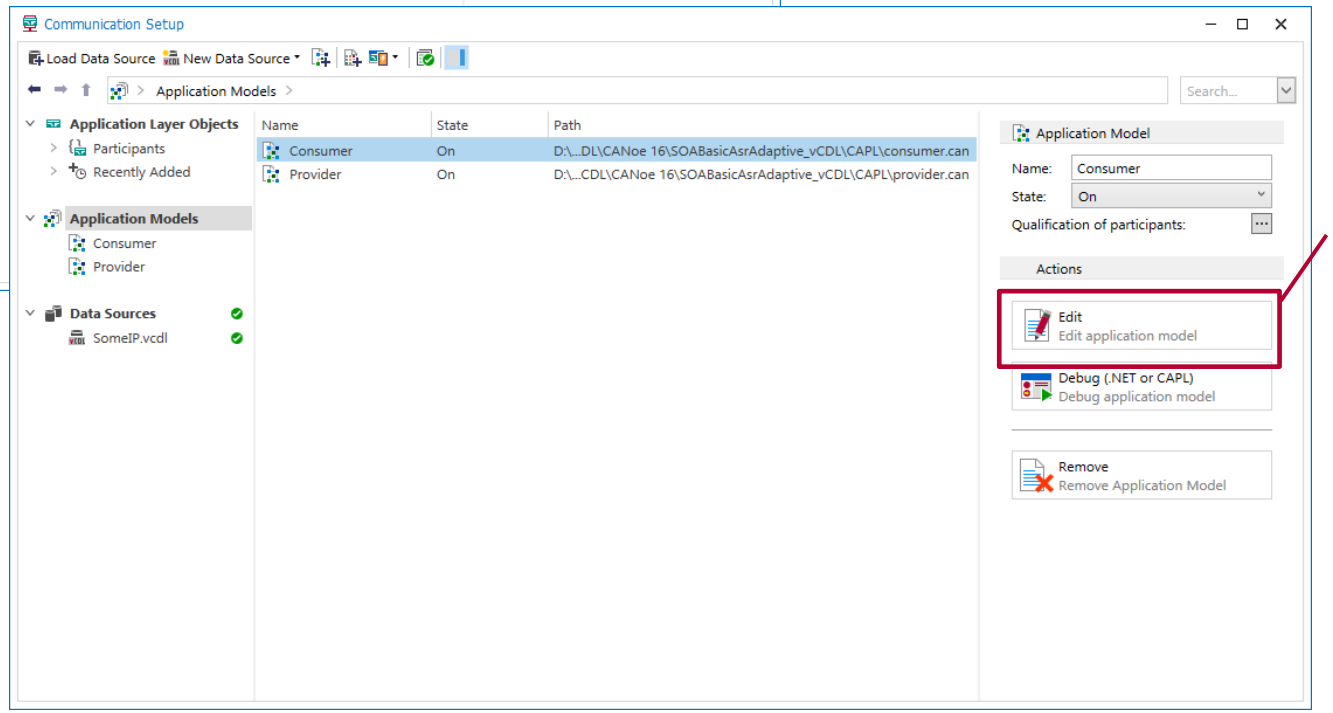
Workflow 1 – Configure Application Models



Step 10

Assign the application models(CAPL, C#)





Step 11

Edit the application models(CAPL, C#)

Workflow 1 – Implement Application Models

consumer.can

```
// Call method 'Add' of service 'Calculator' when key 'c' is pressed
on key 'c'
{
  CommunicationObjects::Calculator[Consumer,Provider].Add.CallAsync(3,6);
}

// This handler is called when the return value of method 'Add' of service 'Calculator' is received
on fct_returned CommunicationObjects::Calculator[Consumer,Provider].Add
{
  // Print result in the Write Window of CANoe
  write("Result of Add method: %f", this.result);
}
```

provider.can

```
on start
{
  // Set timer to cyclic interval of 1000 milliseconds
  setTimerCyclic(stateEventTimer, 1000);
}

on timer stateEventTimer
{
  // Update state event. This will also trigger the transmission of the event
  $CommunicationObjects::Calculator[Provider].State.AddCount = gAddCount;
  $CommunicationObjects::Calculator[Provider].State.SubtractCount = gSubtractCount;
  $CommunicationObjects::Calculator[Provider].State.MultiplyCount = gMultiplyCount;
  $CommunicationObjects::Calculator[Provider].State.DivideCount = gDivideCount;
}

// Implementation for method 'Add'
on fct_called CommunicationObjects::Calculator[Provider].Add
{
  this.result = this.operand1 + this.operand2; // Calculate result
  gAddCount++;                                 // Update counter
  this.ReturnCall();                           // Return result

  // Update CalcResult Field. This will also trigger the transmission of the field notification
  $CommunicationObjects::Calculator.providerSide[Provider].CalcResult = this.result;
}
```

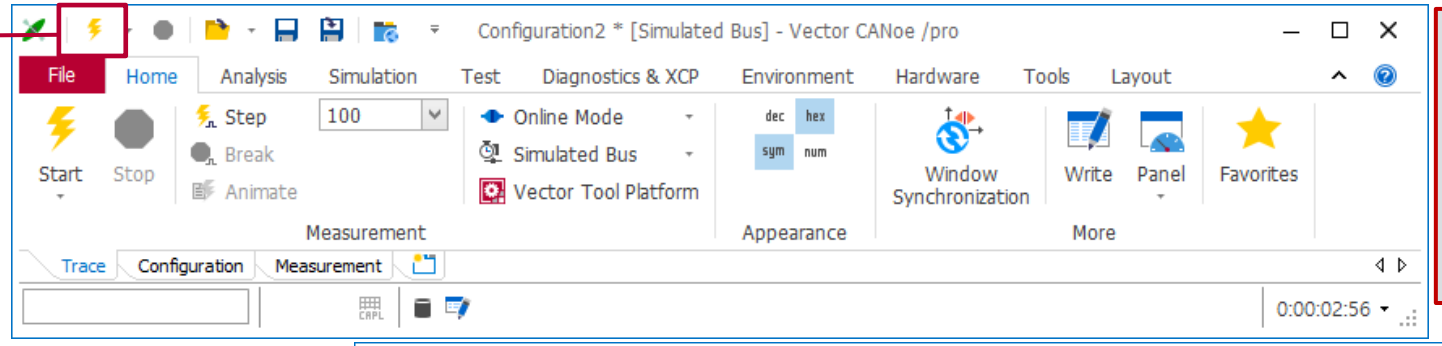
Step 12

Implement the following application models for Provider and Consumer

Workflow 1 – Implement Application Models

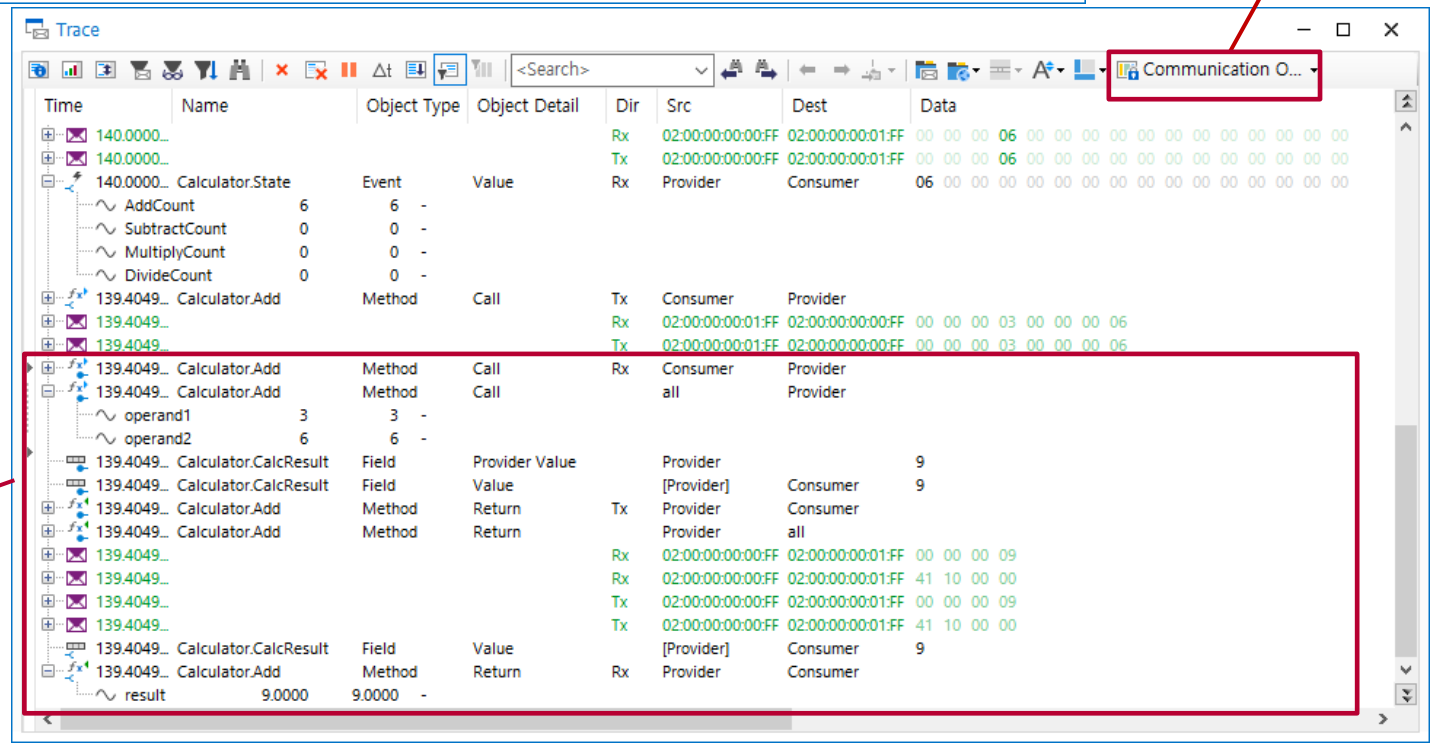
Step 13

Start measurement



Step 14

Switch to Communication Object column view



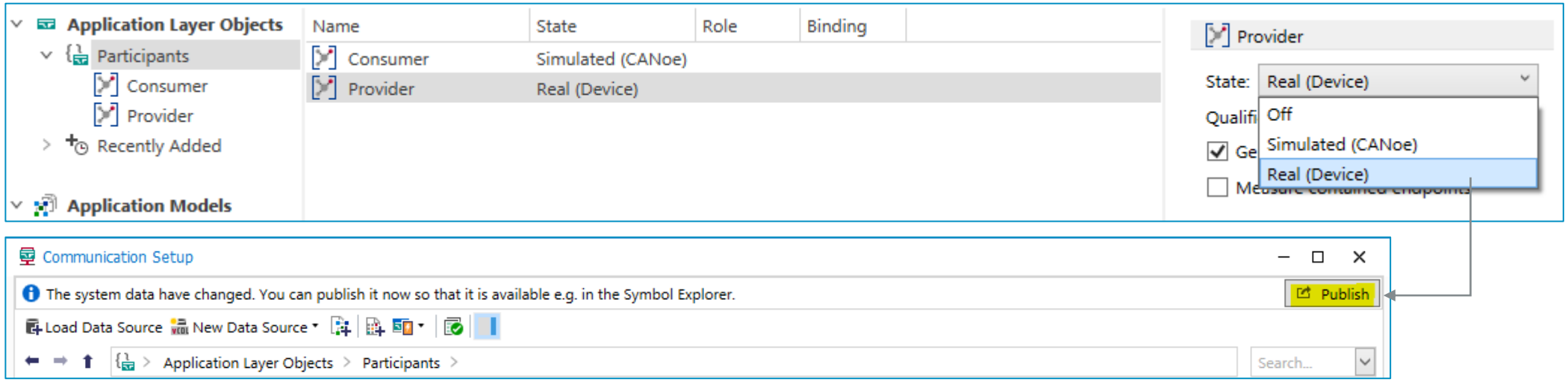
Step 15

Press key „c“

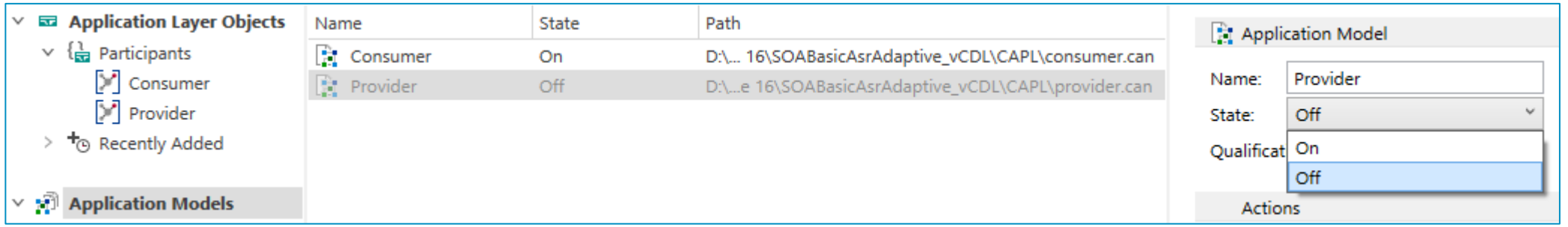
You can observe the method call from Consumer in the trace window

Workflow 2 – Partial Simulation with a Real Target

- If either endpoint Provider or Consumer operates on Real Target:
  - Change step 9: Switch Participant corresponding to Real Target to ‘Real (Device)’ and ‘Publish’



  - Change step 11: Deactivate or remove the corresponding application model

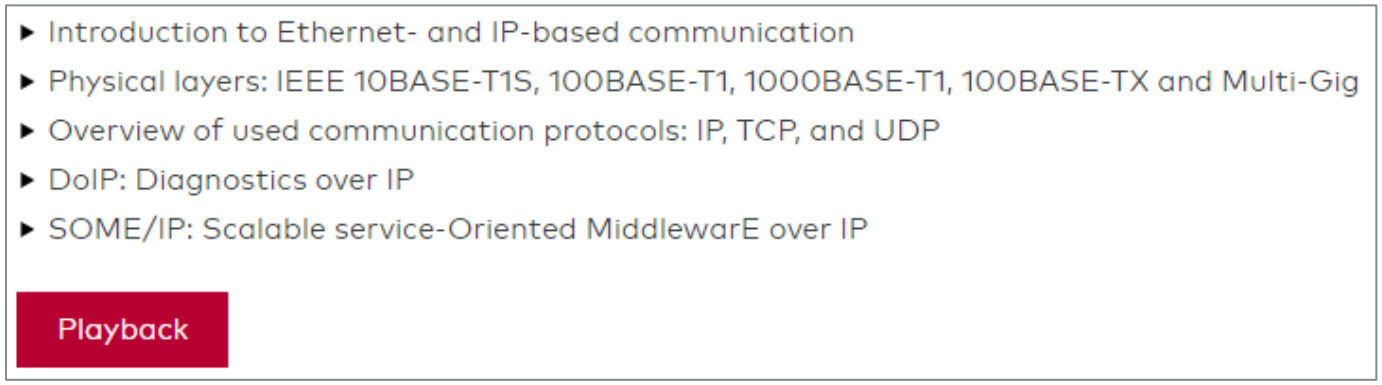


Agenda

- Overview
- Communication Setup
- **Technical References**
- Summary

Video Recording

- Introduction to Automotive Ethernet
  - https://www.vector.com/kr/ko/events/global-de-en/webinar-recordings/2023/introduction-to-automotive-ethernet/

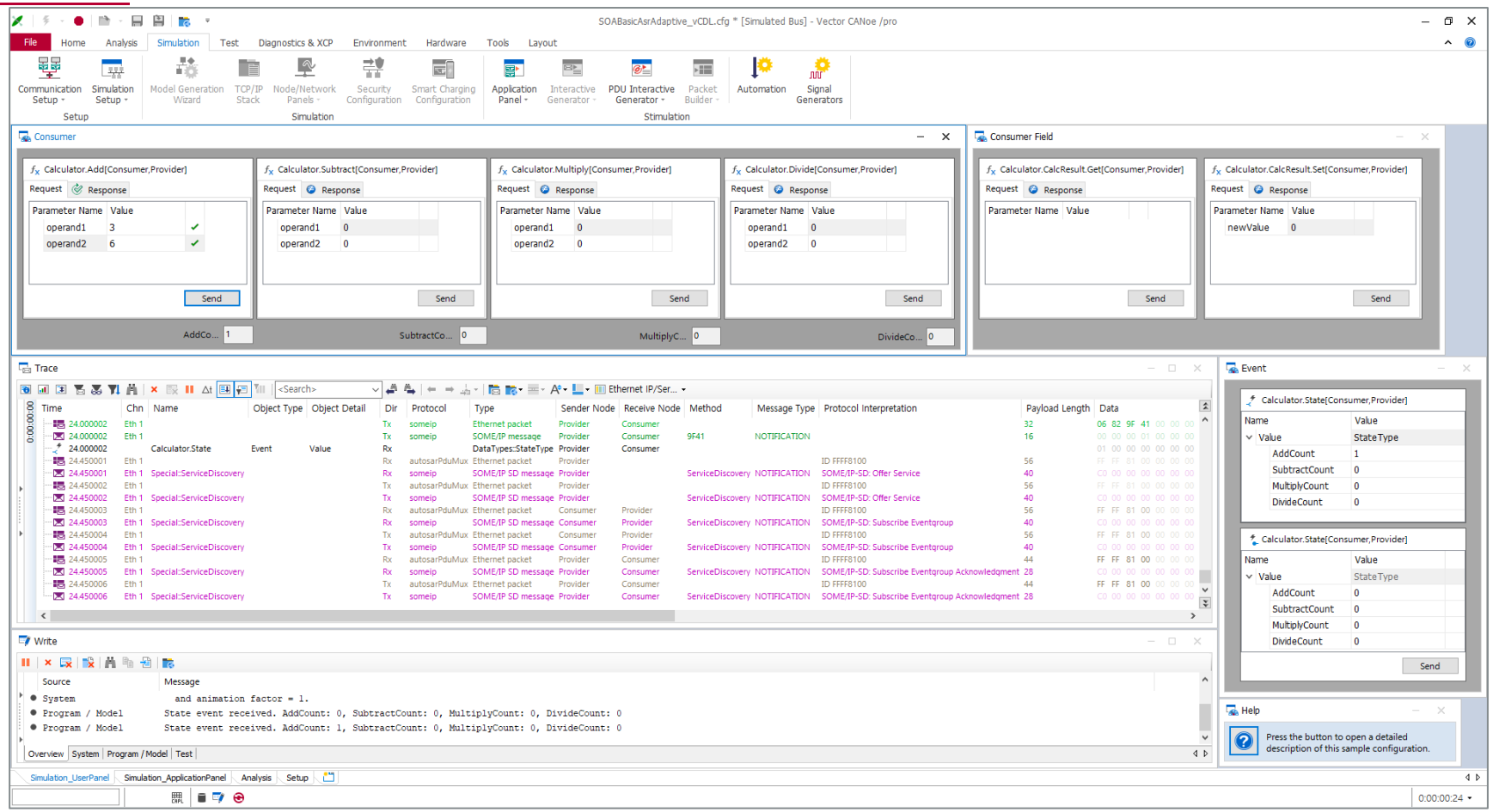


- CANoe for Service-Oriented Architectures(SOA) - Part1. SOA Fundamental
  - https://youtu.be/M_SXOgci1p4



Calculator Example

- CANoe configuration, vCDL, Analysis Features, Simulation Features, CAPL Codes
  - Download Sample Configuration : https://portal.vector.com/shared/068f1303-77de-4eab-8bd0-ca37938c6885

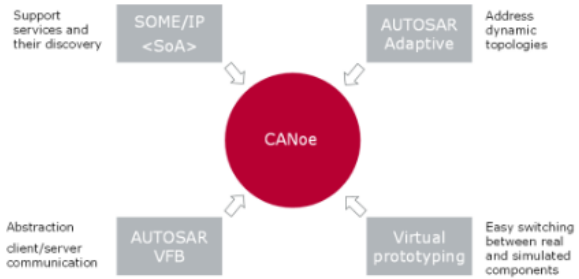


CANoe Help Manual

CANoe Communication Concept Overview

CANoe » Communication Concept

Communication Concept



CANoe as simulation and test environment provides an uniform and cross-network communication concept. This communication concept enables application-related modeling of classic and new communication patterns. The communication concept provides consistent decoupling of logical communication on the application level from protocols of the network level.
This means that CANoe provides a network-independent interface for application models and test scripts. Separately from this, the actual access or transmission is configured via a specific network.

- **Comparison**: How was it before – how is it now?

Concept

- Basics of the Communication Concept
- Service-Oriented Communication
- Binding

Application Layer Objects

- Communication Objects
- Distributed Objects
- Display and Stimulate Application Layer Objects

Configuration

- Communication Setup
  - Load Data Source
  - AUTOSAR Preprocessor
  - CANoe AUTOSAR Converter

Binding

Communication Objects

- Abstract Binding
- SOME/IP Binding

Distributed Objects

- Abstract Binding
- MQTT Binding
- HTTP Binding
- DDS Binding
- IoT Enabler
- Mapping
- CAPL
- C#
- FDX

Serialization

- JSON
- Google Protocol Buffers

Programming

- Programming C#, Python and CAPL
  Description how you can use the objects and the APIs of the communication concept in C#, Python or in CAPL to create, for example application models or to test functionality.
- Vector Communication Description Language (vCDL)
  vCDL allows the CANoe communication concept to be extended by a tool for the simple and text-based configuration of communication objects.

Agenda

- Overview
- Communication Setup
- Technical References
- **Summary**

This document is for …

- Beginners in SOME/IP with Communication Setup

- Help to understand for the Initial Setup
  - SOME/IP concept in CANoe
  - Import Data Sources
  - Import Application Models
  - Write Data Source as a vCDL
  - Simulation with CANoe GUI features
  - Analysis of SOME/IP data
  - Simulation with CAPL programming

  will be covered in other Parts…

- Quick start guide for work of basic SOME/IP simulation, analysis
  - You can extend your knowledge and know-how through CANoe Help Manual!
  - It covers More detailed technical information.

For more information about Vector
and our products please visit

www.vector.com



Author:
Lee, Jaecheol
Vector Korea

© 2023. Vector Korea IT Inc. All rights reserved. Any distribution or copying is subject to prior written approval by Vector. V1.0 | 2023-07-18